Discussion Agenda 06/11/2018 (CGM+Clumps)

Below is the discussion agenda for a joint telecon of AGORA Paper Groups "CGM" and "Clumps" (June 11, 9am PDT/6pm CEST). This telecon will focus on identifying urgent joint action items for the two science-oriented papers, in preparation for the 2018 summer Workshop.

Contents

Check out the important items in red color below!

(last revision: 07/07/2018)

1. Joint Effort to Launch Papers "CGM" and "Clumps"

We decided to make a joint effort to launch both Papers "CGM" and "Clumps" (discussed in detail below) before the August Workshop 2018. After that, each Paper Group will write up its own paper using the same suite of simulations.

	Task Force "CGM"+"Clumps"
GOALS	• Launch the simulations to be used in two Papers "CGM" and "Clumps" by Aug. 10, 2018.
TASK FORCE COORDINATOR	• Santi Roca-Fabrega (UCM)
CODE LEADERS	• Those who will actually carry out the calibrations #1, #2 and the production simulation: **Code** – **Contacts** ART-I – Ceverino/Roca-Fabrega ENZO – Hummels/Kim GADGET – Nagamine/Shimizu GASOLINE – Shen GEAR – Hausammann GIZMO – Dong/Lupi RAMSES – Roca-Fabrega
MILESTONES	• Calibration Step #1 (Jun. 12 - Jun. 30, 2018): "Favorite" feedback calibration step #1 described in the Paper "CGM" Workspace (i.e., isolated disk IC, tweak each group's "favorite" feedback recipe to give a stellar mass of M_* ~ 1e9 Ms produced in the first 1 Gyr, meanwhile check if your old run with the "common" feedback recipe produced a similar stellar mass). Do as many iterations as possible to finish this calibration. Please send your final entry to Santi as soon as possible, no later than June 30 (note: Grackle 2.1 users, see note 1 below). • Calibration Step #2 (Jul. 1 - Jul. 31, 2018): "Favorite" feedback calibration step #2 described in the Paper "CGM" Workspace (i.e., cosmological IC, check each group's "common" and "favorite" feedback recipes to give a similar stellar mass at z=4, check other runtime parameters in this Workspace such as gravitational softening). Please carry out as many iterations as possible to finish this calibration. Please send your final entry to Santi as soon as possible so we can start comparing the data across the codes, no later than July 31.

[Note 1] Using Grackle v2.1 implies that an updated default value for solar abundance Z_{sun} = 0.01295 is used (as opposed to 0.02041, a default in Grackle v2.0 or below). More discussion can be found by searching "Grackle" in the **Workspace** of our first disk comparison paper (previously known as **Paper "4"**). Note that this issue was not fully resolved in our previous paper, leading to a discrepancy in rho-T diagram. Therefore, for our new Papers "CGM"/"Clumps" and all preceding calibrations, we will use the common solar metallicity value of Z_{sun} = 0.02041 for all codes. To set a new solar abundance in Grackle v2.1 (used by GADGET/GASOLINE/GIZMO groups as of 2018), the only thing you have to do is to manually change the default value of Z_{sun} during the Grackle initialization. Simply add this line after the call to the set_default_chemistry_parameters routine in the Grackle initialization: grackle_data.SolarMetalFractionByMass = 0.02041;

2. Paper Group Progress Report

We are making progress, chiefly, in the six different fronts below.

(1) Science-oriented Projects

<table>
<tr><th></th><th>Paper "CGM"</th><th>Paper "Clumps"</th><th>Paper "Metallicity"</th></tr>
<tr><td>GOALS</td><td>• Run ~1e12 Ms halo to study absorption lines</td><td>• Run ~1e12 (~1e13) Ms halo to study SF clumps at high z</td><td>• Run ~1e12 Ms halo to study metal distribution</td></tr>
<tr><td>LEADERS</td><td>• Cameron Hummels (Caltech)
• Santi Roca-Fabrega (UCM)</td><td>• Daniel Ceverino (Heidelberg)
• Alessandro Lupi (IAP)</td><td></td></tr>
<tr><td>PARTICIPANTS</td>
<td>• Code representatives:
<table>
<tr><th>Code</th><th>Contacts</th></tr>
<tr><td>ART-I</td><td>Dekel, Mandelker, Primack, Roca-Fabrega, Strawn</td></tr>
<tr><td>CHANGA</td><td>(Quinn)</td></tr>
<tr><td>ENZO</td><td>Hummels, Kim, O'Shea, Wise</td></tr>
<tr><td>GADGET</td><td>Nagamine, Shimizu</td></tr>
<tr><td>GASOLINE</td><td>Madau, Shen, Wadsley</td></tr>
<tr><td>GEAR</td><td>Hausammann, Revaz</td></tr>
<tr><td>GIZMO</td><td>Dong, Hopkins, Hummels</td></tr>
<tr><td>RAMSES</td><td>Roca-Fabrega, Kretschmer, Teyssier</td></tr>
</table>
(Code leaders as of Jun. 2018)
- ICs help: Buehlmann, Hahn, Simpson
- Analysis help: Chakrabarti, Fumagalli, Hummels, Kim, Madau, Roca-Fabrega, Smith, Turk

• Working Groups engaged:
- III (common cosmo ICs)
- IV (common analysis)
- V (subgrid physics)
- IX (char. of cos. galaxies)</td>
<td>• Code representatives:
<table>
<tr><th>Code</th><th>Contacts</th></tr>
<tr><td>ART-I</td><td>Dekel, Ceverino, Mandelker, Primack</td></tr>
<tr><td>CHANGA</td><td>(Quinn)</td></tr>
<tr><td>ENZO</td><td>O'Shea</td></tr>
<tr><td>GADGET</td><td>Nagamine, Shimizu</td></tr>
<tr><td>GASOLINE</td><td>Mayer, Wadsley</td></tr>
<tr><td>GEAR</td><td>Hausammann, Revaz</td></tr>
<tr><td>GIZMO</td><td>Lupi, Hopkins</td></tr>
<tr><td>RAMSES</td><td>Agertz</td></tr>
</table>
(Code leaders as of Jun. 2018)
- ICs help: Buehlmann, Hahn
- Analysis help: Chakrabarti, Kim, Turk

• Working Groups engaged:
- III (common cosmo ICs)
- IV (common analysis)
- XI (high-z galaxies)</td>
<td>• Code representatives:
<table>
<tr><th>Code</th><th>Contacts</th></tr>
<tr><td>GADGET</td><td>Chakrabarti, Lewis</td></tr>
<tr><td>GEAR</td><td>(Revaz)</td></tr>
<tr><td>RAMSES</td><td>(Pallottini, Teyssier)</td></tr>
</table>
- Analysis help: Kim

• Working Groups engaged:
- IV (common analysis)
- IX (char. of cos. galaxies)
- XII (interstellar medium)</td></tr>
</table>

	– X (outflows)		
PROGRESS MADE OR BEING MADE	• Visit **Paper's Workspace** for the latest updates. • See the joint effort to launch both Papers "CGM" and "Clumps" in Section 1 above. • Trident is now fully functional with "demeshened" version of yt (i.e., **yt-4.0** branch) as of Jun. 2018. (see also **this link** about Trident v1.1 as of Nov. 2017) • Hummels has provided an **ipython notebook** that explains how to use yt+Trident on datasets from various codes.	• Visit **Paper's Workspace** for the latest updates. • See the joint effort to launch both Papers "CGM" and "Clumps" in Section 1 above.	• Visit **Paper's Workspace** for the latest updates. • Will piggyback on the planned Paper "CGM" runs and evolve them further down to z=0.
OTHER LINKS	• **Paper Workspace** • **Trident download and documentation** • **Trident method paper (Hummels et al. 2016)**	• **Paper Workspace**	• **Paper Workspace**

Cf. other science-oriented project ideas: see **this link**

(2) Previous Projects

	Paper "DM-Highres"	**Paper "NoSubgrid"**	**Paper "Disk-II"**
GOALS	• Run Proof-of-concept-DM (~1e11 Ms halo) at higher res	• Run Proof-of-concept-NOSUBGRID (~1e11 Ms halo)	• Compare simulation codes using common disk ICs
LEADERS	• Oliver Hahn (OCA, Nice) • Christine Simpson (HITS)	• Nick Gnedin (Fermilab) • Piero Madau (UCSC) • Britton Smith (SDSC)	• Santi Roca-Fabrega (HUJI)
PARTICIPANTS	• Code representatives:	• Code representatives:	• Code representatives:

Paper "DM-Highres" code representatives:

Code	**Contacts**
ART-I	Ceverino, Klypin, Primack
ART-II	Gnedin, Feldmann, Li, Semenov (Leitner)
CHANGA	Quinn, Wallace
ENZO	Hummels, Kim, Simpson, Wise
GADGET	Choi, Nagamine, Onorbe, (Rocha)
GADGET-AFS/CFS	(Hobbs)
GASOLINE	Madau, Shen, (Pillepich,

Paper "NoSubgrid" code representatives:

Code	**Contacts**
ART-I	Ceverino, Klypin
ART-II	Gnedin, Feldmann, Leitner
CHANGA	Quinn
ENZO	Hummels, Kim, Simpson, Smith, Turk, Wise
GADGET	Choi, Nagamine
GADGET-SPHS	(Hobbs)
GASOLINE	Keller, Madau, Mayer, Shen, Wadsley
GEAR	Revaz

Paper "Disk-II" code representatives:

Code	**Contacts**
ART-I	Arraki, Ceverino, Klypin, Mandelker, Snyder
ART-II	Feldmann, Gnedin, Leitner, Li, Semenov
CHANGA	Quinn, Wallace
ENZO	Abel, Butler, Butsky, Forbes, Hummels, Kim, O'Shea, Simpson, Turk, Wise
GADGET	Choi, Nagamine, Chakrabarti, Shimizu,

	Zolotov) GCD+ — Kawata GIZMO-AFS — Garrison-Kimmel, Lupi, Hopkins, Onorbe PKDGRAV2 — (Guedes, Kuhlen) RAMSES — Hahn, Roca-Fabrega, Teyssier (Code leaders as of Aug. 2016) - Others interested: Dekel, Feldmann, Kawata, Pallottini, Robles	GIZMO — Lupi, Thompson RAMSES — Agertz, Butler, Teyssier (Grackle implementers acknowledged in Paper "4")	(Todoroki) GADGET-SPHS — (Hobbs) GASOLINE — Benincasa, Keller, Madau, Mayer, Shen, Wadsley GEAR — Revaz GIZMO — Lupi, Hopkins, Thompson RAMSES — Agertz, Butler, DeGraf, Teyssier, Nickerson (Code leaders as of Aug. 2016 for Paper "4" before)
PROGRESS MADE OR BEING MADE	• Visit Paper's Workspace where Simpson, Hahn and others have been updating the progress. • We will plan to have 5-code comparison. As of Aug. 2017, 4+ participating groups have completed and submitted the production-level results (~100 pc resolution) to NERSC. • We may focus on the (survival of) substructures. This subset of topics could potentially to be led by Jiang as a smaller paper. • Consistent tree - python interface has been built by Britton Smith (ytree).	• Code groups such as ART-I and GIZMO have carried out cosmological simulations with various AGORA ICs, even including sophisticated subgrid physics.	• Visit Paper's Workspace where the results for disk comparison are compiled (including those presented in our first disk comparison paper - previously known as Paper "4"). The agora-analysis-script repository has been accordingly updated. • Analysis items that were not covered in our first disk comparison paper (see **this link**) will be considered in the next paper, Paper "Disk-II". • A subset of topics could be discussed in the "Metal Diffusion" or "Secular evolution" paper (see this link).
OTHER LINKS	• Progress report in Paper's Workspace and/or in WG III's Workspace. • MUSIC Google group • ytree documentation	• GRACKLE Google group • GRACKLE download & documentation • GRACKLE method paper (Smith et al. 2017) • AGORA API for GRACKLE by Nick Gnedin	• Progress report in Paper's Workspace and/or in WG V's Workspace. • yt-AGORA Google group

3. Recaps

(1) AGORA SMBH Prescription for Accretion and Feedback:

- Please visit **this link**.

(2) Strategy to Store Simulation Outputs:

- Please visit **this link**.

(3) How to request an account on NERSC:

- If you represent a code group but don't have a NERSC user account, please visit **this page**. Select "standard" account and find the repository name "agora". Joel Primack will add your name to the group when approved.

(4) Your Own yt Installation on NERSC:

- Please visit **this link**.

(5) yt-Sunrise Pipeline for AGORA on NERSC / Working Installation of Sunrise as Docker Image:

- Please visit **this link**.

(6) Mini-workshop Opportunities:

- Mini-workshop for individual paper groups to write papers: KIPAC@Stanford is willing to provide funding opportunities for working group leaders to organize mini-workshops (~10 people). An example was the **yt-AGORA mini-workshop** at UCSC in Mar. 2014.

(7) Other Useful Links:

- Quick links to the important pages in AGORA Workspace, including internal policies: **here**
- Internal policy regarding the disclosure of AGORA simulation results: **here**
- Collection of softwares relevant to AGORA effort: **here**

4. Your Feedback

Let us hear anything AGORA, including the scientific and organizational effort currently being made. Voice your opinion during the web conference! Feel free to share your thoughts in the comments section below, or on the **mailing list**. You can also directly contact the **project coordinator** anytime!